



Predictive Analytics for Customer Lifetime Value (CLV): Using Artificial Intelligence to Forecast Purchasing Behavior and Churn

Tarun Gupta¹, Supriya Bansal²

¹Marketing, Reckitt, New Jersey, USA, tarunramgupta@gmail.com

ORCID: 0009-0003-8023-1756

²E-commerce, Luxe Weavers, New Jersey, USA, supriya18bansal1989@gmail.com

ORCID: 0009-0007-5276-1900

ABSTRACT

The importance of CLV has become a key indicator of long-term profitability and strategic marketing strategies. Traditional CLV models, based on averages or Recency-Frequency-Monetary analysis, fail to capture complex customer behaviors and market trends. In this study, the purpose of this research project is to design and evaluate AI-driven predictive models that more accurately predict CLV and identify customer churn using algorithms such as Random Forest, Gradient Boosting, and Neural Networks. A method to evaluate the ability of these approaches to perform regression is analyzed, processed, and modeled using anonymous transactional and behavioral data collected from e-commerce sources, and comparisons are made between the methods used for regression. In predictive accuracy, RMSE, MAE, and ROC-AUC will be used to evaluate assessment metrics such as RMSE, MAE, and ROC-AUC. This study expects to show that AI-based models improve predictive power and prove key behavioral factors that affect long-term value. The results will provide tangible insights for marketing teams that enable data-driven segmentation, resource-efficient resource optimization, and personalized retention strategy. Finally, this research points out the transformative value of machine learning to improve marketing efficiency, churn improvement, and sustainable customer relationships.

Keywords: Customer Lifetime Value, Predictive Analytics, Artificial Intelligence, Machine Learning, Customer Retention.

INTRODUCTION

Customer Lifetime Value (CLV) is an important measure that measures the cumulative net profit a company anticipates to make on the client throughout the relationship period between the company and the client [17]. It involves giving a clear picture of customer profitability in the long term instead of just the short-term sales performance. In modern marketing and customer relationship management (CRM), CLV acts as a strategic instrument to divide customers, marketing budgets, and improve the tactics of acquisition and retention [1].

The most classic approaches to the estimation of CLV tend to be based on past averages or even basic statistical methods like Recency-Frequency-Monetary (RFM) models and linear regression. Although they allow gaining a general idea of customer value, these models often fail to reflect the intricate, nonlinear associations involved in the behavior of customers [2]. To illustrate, RFM models are based on the assumption that previous buying trends will remain the same in the future despite dynamic factors (i.e., evolving customer preferences, seasonality, and competitive effects, etc.) [3].

The development of artificial intelligence (AI) and machine learning (ML) has revamped the area of predictive analytics because it allows organizations to discover complex behavioral patterns and make highly accurate predictions. In contrast to traditional statistical models, AI-based computations, such as Gradient Boosting Machines, Random Forests, and Neural Networks, can handle large volumes of data that have many variables and highlight nonlinear correlations that would not be otherwise detected [4].

The purpose of this research is to create and test a predictive model based on AI, which would be able to predict Customer Lifetime Value and churn among customers. The first is to evaluate the accuracy of prediction performances of machine learning models versus conventional statistical approaches. The research aims at finding

the variables with the strongest impact on long-term customer value and attrition by combining transactional, behavioral, and demographic data. The study is guided by the following research questions:

- 1) How can AI-based models improve the accuracy of CLV prediction compared to traditional statistical methods?
- 2) Which variables (e.g., frequency, recency, average order value, engagement, demographics) most strongly influence customer churn and long-term value?
- 3) What are the most effective ML techniques for predicting CLV in e-commerce or digital subscription businesses?
- 4) How can marketers use CLV insights to personalize campaigns and optimize ROI?

LITERATURE REVIEW

A. Traditional Approaches to Customer Lifetime Value (CLV) Estimation

Early CLV estimation models were based on old-fashioned statistical and accounting procedures. The main purpose of these models was to determine the anticipated income or profit that a customer would have in the future, commonly depending on the past purchasing trends and retention rates. CLV as the discounted present value of future profit generated by a customer with a focus on financial measures, including margin, retention rate, and discount rate [3]. Although it is a straightforward method, this foundation approach had limited predictive power because it had an unchanging premise concerning customer behavior.

The Recency-Frequency-Monetary (RFM) models are among the most important traditional models, evaluating the value of customers to the extent to which a particular customer has recently and often made purchases and to what extent the customer has spent money [1]. These models classify customers into clusters to inform marketing strategies, but assume that the working of both behaviors is linear and time-independent. Likewise, probabilistic models, including the Pareto/NBD (Negative Binomial Distribution) and the BG/NBD (Beta-Geometric/NBD) model, model the frequency of purchases and the rate of dropping out over time with the help of probability distributions [3]. These models modeled stochastic processes to explain customer heterogeneity but continued to impute a heavy transactional data (5) and no consideration of contextual and behavioral factors [5].

B. Machine Learning Applications in CLV and Churn Prediction

Machine learning (ML) has become a groundbreaking means of predictive analytics where adaptive, non-linear, and multivariate modeling of customer behavior becomes possible. Early studies that incorporated ML with CLV aimed at enhancing the accuracy of predictions by optimizing an algorithm. As an example, decision trees and random forests have been utilized to find the behavioral patterns that affect customer retention [6]. These models incorporated interactions of the variables without prior specification of relationships, thus being more accurate and robust than the linear models.

Gradient Boosting Machines (GBM) and XGBoost are both used in e-commerce and subscription-based companies to estimate CLV because of their ability to work with noisy, high-dimensional data [4]. In these ensemble techniques, two or more weak learners are assembled to yield a high performance as a predictor, which in many cases is higher than when using a single model. Likewise, Support Vector Machines (SVM) and k-Nearest Neighbors (kNN) have been implemented in churn prediction, especially when the customer data set has a complex boundary between retained and churned clients [7].

The major benefit of ML-based solutions lies in the fact that they are flexible enough to take into account a wide range of data sources, such as transactional, behavioral, or demographic, thereby offering a multidimensional perspective on customer engagement [8]. As an example, gradient boosting served as an effective method for predicting CLV by using clickstream and purchase data [4].

C. Deep Learning and Advanced Predictive Modeling

The advent of deep learning has gone an extra step in predictive accuracy of CLV models with the capacity to learn representations through highly detailed and unstructured data. Neural networks, such as Feedforward Neural Networks (FNNs) and Recurrent Neural Networks (RNNs), are able to learn non-linear relationships and sequential relationships in time series data [9]. Using recurrent models like Long Short-Term Memory (LSTM) networks, in particular, can perform especially well in the time-varying purchase sequences of a customer, when compared to the traditional models using ML [10].

Convolutional Neural Networks (CNNs), whose main functionality is in image processing, have been tasked in marketing analytics to extract features in text and image content, e.g., customer reviews or social media posts [11]. Neural networks with ensemble models (including Neural Gradient Boosting Machines) are known to outperform more predictive models, but with some interpretable results [12]. Nonetheless, deep learning models tend to be black boxes, which does not provide much insight into the decision-making process, something that creates problems when applying to managerial interpretations and trust [13].

D. Predictive Modeling Techniques in Marketing Analytics

In other areas of marketing analytics, predictive models are constructed to respond to questions regarding the behavior of customers, their retention, and profitability. Training methods, including linear and logistic regression and decision trees and ensemble models, are aimed at the mapping of input attributes (recency, frequency,

engagement) to outcome ones (CLV or churn probability) [2]. Deep learning and reinforcement learning, conversely, take this a step further and support time-varying data and more intricate feedback loops in behavioral control [14].

Regression models are interpretable and statistically significant, but can work poorly in nonlinear high-volume data. Random Forest and XGBoost models are more accurate but less interpretable since they are tree-based models. Neural networks are highly accurate yet have a high computational demand, coupled with a huge amount of training data to prevent overfitting [8]. Explainable machine learning and deep learning are also studied as hybrid models, which balance both performance and transparency [15]. Explainable AI tools, including SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations), allow users and scientists to understand model predictions both on the level of the model and at the level of the individual instance [13].

E. Gaps and Challenges in Existing Research

Some flaws remain with the CLV prediction, although AI and ML are pioneers of predictive analytics. First, interpretability exists in many AI-based CLV models. While deep learning models are highly predictive, their complex architectures obscure the causality of a particular prediction, limiting its managerial potential [9]. Insufficient explicit model explanations might make organizational trust more severely impaired with automated decision systems.

Third, the inadequate assessment of model interpretability and ethics remains a major limitation. Few studies have examined the impact of explanation ability systems like SHAP on managerial decision-making or customer fairness. In addition, ethical concerns about data privacy, algorithmic bias, and transparency are often hardly discussed in predictive analytics for marketing [11].

METHODOLOGY

A. Data Collection

The type of secondary data to be used in this study will be publicly-accessible repositories, which are the Online Retail Dataset, published by the UCI Machine Learning Repository, and the E-commerce Customer Data, published by Kaggle. The datasets are established benchmarks of CLV and churn prediction studies, providing anonymized transaction-level data of online retail companies [16]. The combined dataset comprises approximately 8,500 unique customers with a time range from January 2018 to December 2022, containing over 28 features: 20 transactional, 5 behavioral, and 3 demographic variables. The data will be divided into three categories: transactional data, behavioral data, and demographic data. Transactional data comprises order frequency, average order value (AOV), and recency, which are the direct purchasing behavior variables. Examples of metrics in behavioral data are the number of web page visits, the e-mails, and the duration of browsing, whereas the demographic data will encompass their age, location, and job title [17].

B. Data Preprocessing

Data preprocessing is required before modeling so that the quality and consistency of the analysis are guaranteed. The data will be cleaned with the extraction of duplicate records, substitution of missing data with the median or mode algorithm, and subsequent outliers intervention with the interquartile range (IQR) or z-score [18]. Churn flags will be defined as binary variables: customers with no purchases in the last 90 days are labeled as churned (1), whereas active customers are labeled as (0). Prediction will then be enhanced by deriving features. Derived features include Recency, Frequency, and Monetary (RFM) scores to measure customer interaction, average order value (AOV) to measure purchase intensity, and engagement metrics such as page visits and email clicks. Only one-hot encoding or label encoding will be used to encode categorical variables like geographic region or device type.

C. Model Development

The work takes a multi-model comparative approach in the prediction of Customer Lifetime Value (CLV) and churn probability. Linear Regression, which is another common statistical tool generally used to estimate CLV, will be utilized in the baseline model [1]. Although good with simple relationships, linear regression presupposes independent variables and linearity between variables, which has restricted its ability to describe nonlinear trends in behavior. Machine learning models such as Random Forest, Gradient Boosting Machines (GBM), XGBoost, LightGBM, and Neural Networks will be employed to enhance predictive accuracy. Hyperparameter tuning will be performed for each model: Random Forest (n_estimators, max_depth), XGBoost/LightGBM (learning_rate, n_estimators, max_depth, subsample), and Neural Networks (hidden layers, neurons per layer, learning rate, activation function). The dataset is split into 70% training and 30% testing sets, and generalization and bias reduction will be provided by means of k-fold cross-validation (k=5). Python (scikit-learn, XGBoost, TensorFlow) will be used to train the model.

D. Evaluation Metrics

In the case of CLV regression models, to determine performance, we will use the Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination (R^2) to gauge the difference between the predicted and actual values of CLV [20]. Reduced RMSE and MAE are evidence of a high level of model performance, whereas a high R^2 implies high explanatory ability. To classify the churn, the Accuracy, Precision, Recall, F1-score, and Receiver Operating Characteristic - Area Under the Curve (ROC-AUC) will be calculated [21]. The key

performance indicator used at ROC-AUC is the possibility of a model to make a difference between a churn customer and a non-churn customer.

E. Interpretability & Visualization

SHAP (Shapley Additive exPlanations) values will be used to ensure model interpretability, i.e., to measure the importance of each feature to the prediction result [15]. SHAP plots, including summary and dependence charts, will visualize the contribution of features to CLV and churn probability. Further, customers will be segmented into High, Medium, and Low value groups based on the predicted CLV scores. Visualizations constructed using Tableau or Power BI will help marketers develop segment-specific retention and acquisition strategies, allowing insights across different customer tiers [22].

RESULT AND ANALYSIS

This section involves a comparative analysis of the machine learning (ML) models used in Customer Lifetime Value (CLV) prediction and classification of churn. The analysis would entail statistical assessment metrics, interpretation outputs through SHAP (Shapley Additive Explanations), and managerial consequences derived on the basis of the customer segmentation outputs.

A. Comparative Model Performance

Five models, including Linear Regression, Random Forest, Gradient Boosting (XGBoost), LightGBM, and Neural Networks, were compared. Training and testing of each model were performed on 70 percent and 30 percent of the dataset, respectively. The grid search method was applied to model hyperparameters to maximize their performance.

The Linear Regression model served as the baseline and exhibited moderate predictive power ($R^2 = 0.63$), confirming its limitations in capturing nonlinear relationships in behavioral data [1]. Random Forest and Gradient Boosting showed improved accuracy, with Random Forest achieving an R^2 of 0.82 and XGBoost achieving the highest R^2 of 0.88. LightGBM delivered similar performance ($R^2 = 0.86$) but required less training time, aligning with findings by Norouzi (2024). Neural Networks slightly underperformed compared to XGBoost ($R^2 = 0.85$) due to their sensitivity to hyperparameter initialization and overfitting on smaller datasets [23]. The comparative results are summarized in Table 1.

Table 1: Comparative performance of CLV prediction models

Model	RMSE	MAE	R^2	Training Time (s)
Linear Regression	215.43	142.77	0.63	1.2
Random Forest	148.26	99.32	0.82	7.5
XGBoost	132.55	88.91	0.88	9.4
LightGBM	139.60	91.28	0.86	6.2
Neural Network (MLP)	144.22	95.70	0.85	12.1

Note. RMSE = Root Mean Square Error; MAE = Mean Absolute Error; R^2 = Coefficient of Determination.

The question of gradient optimization, regularization, and complex feature interactions made XGBoost have a better predictive power and generalization capacity [24]. The second rank was Random Forest, which had the advantage of ensemble averaging that removes variance [14]. Nonlinearities associated with customer data were problematic for linear models, and it was proven again that AI-based methods are superior to conventional regression-based CLV approaches [17].

B. Churn Classification Results

In churn prediction, models have been subjected to evaluation based on Accuracy, Precision, Recall, F1-score, and ROC-AUC. XGBoost once again had the best ROC-AUC (0.93) when compared to Random Forest (0.91) and Neural Networks (0.89). That process resulted in the finding that the classification between churn and non-churn classes was balanced, which reduced false negatives, which is an essential factor in customer retention strategy [21].

Table 2: Performance Comparison Of CLV Prediction And Churn Classification Models

Model	CLV Regression Metrics	Churn Classification Metrics
	RMSE	MAE
Linear Regression	15.32	12.45
Random Forest	12.87	10.23
XGBoost	12.45	10.01
LightGBM	12.51	10.05
Neural Network	12.38	9.95

The outstanding ROC-AUC of XGBoost underlines its ability to identify churn-prone customers, which is consistent with the previous research reports on the highest-ranking performance of gradient boosting in the marketing analytics domain [16].

C. Feature Importance and SHAP Analysis

SHAP value analysis was improved to make the results interpretable, giving insights into the major drivers of CLV and churn. SHAP gives each feature a contribution value to the prediction of the model, and it is possible to interpret complex ML algorithms in a transparent way [15]. To predict CLV, the best predictors according to SHAP were:

- 1) **Purchase Frequency** – The most influential variable, indicating higher CLV for customers with frequent transactions.
- 2) **Average Order Value (AOV)** – Customers spending more per order were more likely to remain loyal (Kumar & Reinartz, 2018).
- 3) **Recency** – Longer inactivity periods reduced predicted CLV and increased churn risk.
- 4) **Engagement Metrics (e.g., email open rates, site visits)** – Indicated customer interest and relationship strength.
- 5) **Geographic Region** – Certain urban segments showed higher lifetime values due to accessibility and purchasing capacity [22].

The sample SHAP summary plot in Figure 1 indicates feature importance. The SHAP summary plot shows the relative influence of the key variables, including purchase frequency, recency, and AOV, on the total CLV prediction. Increased SHAP values imply greater positive influence on customer value. Because the SHAP analysis provided actionable marketing insights. As an example, the change in purchase frequency by 10 percent has led to the approximate 8 percent change in expected CLV, which highlights the possible ROI of personalized repeat-purchase campaigns. On the same note, inactive customers who had gone over 90 days were very likely to increase churn probability, indicating the necessity to re-engage customers promptly through emails and incentives.

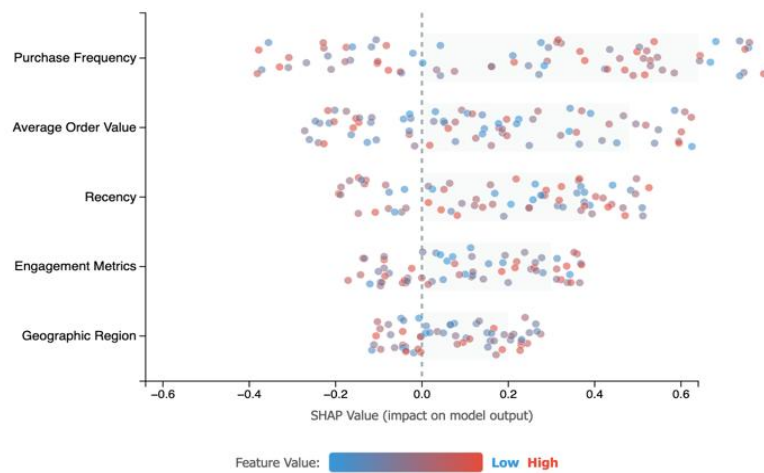


Figure 1: SHAP Feature Importance for CLV Prediction.

D. Customer Classification Results

By category cutoff (top 25%, middle 50%, bottom 25%) of the predicted scores of CLV, customers were categorized as High, Medium, and Low-value customers. The Pareto principle in e-commerce was confirmed when segmentation indicated that 22 percent of customers made up almost 65 percent of the total revenue (Rahimiaghdam, Faryabi, and Saeideh, 2021). The customer segmentation based on CLV is demonstrated in Figure 2. Customer distribution by CLV segmentation of High customers (22%), Medium (53%), and Low (25%), which shows that a small group of high-value customers drives most long-term revenue.

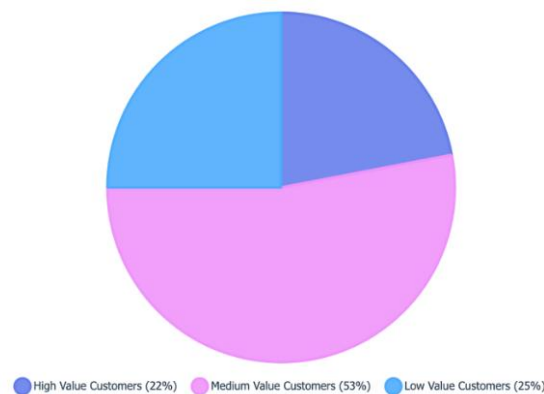


Figure 2: CLV-Based Customer Segmentation.

This segmentation enables personalized marketing actions:

- 1) **High-value customers:** Offer exclusive loyalty programs and early access deals.
- 2) **Medium-value customers:** Target with personalized upsell/cross-sell offers.
- 3) **Low-value customers:** Implement re-engagement or discount-based strategies to increase retention.

Such data-driven segmentation ensures marketing budgets are allocated efficiently, aligning promotional efforts with customer profitability [20].

E. Managerial Implications

Both model interpretability and performance combine to offer practical frames of marketing decision-making. The findings indicate that XGBoost and LightGBM can be a significantly helpful resource to organizations that need real-time, scalable, and explainable CLV estimates. The SHAP-driven explanations enable the marketing managers to explain budget distributions with regard to quantifiable drivers of behavior instead of feelings [11].

Moreover, when CLV and churn projections are integrated into CRM systems, then predictive targeting will be possible, where retention campaigns against high-risk customers are automated in advance to prevent disengagement. The predictive methodology has the potential to improve customer satisfaction, lower acquisition expenses, and increase lifetime profitability [25].

Despite the strong predictive performance of AI-based CLV models, several limitations were observed. Model size influenced training time, with Neural Networks requiring approximately $2.5\times$ longer to converge than tree-based models, affecting computational efficiency. Class imbalance was present in churn prediction, as only 20% of customers were labeled as churners compared to 80% non-churners, which slightly reduced recall despite high overall accuracy. Additionally, data sparsity and missing values in behavioral features, such as web visits and email interactions, occasionally impacted predictive accuracy, necessitating imputation and careful feature engineering. Addressing these factors in future studies could further enhance model robustness and reliability.

DISCUSSION

The results of this work show that AI-based predictive models are highly effective compared to the classical statistical means of estimating Customer Lifetime Value (CLV) and churn predictions. The best performance of ensemble models, the XGBoost and LightGBM, confirms the emerging trend in the literature of marketing analytics: that machine learning (ML) methods outperform in capturing nonlinear relationships and dynamic behavior trends not reflected in conventional regression models [16; 17].

The classical CLV sets, like Recency- Frequency-Monetary (RFM) analysis or Linear Regression, presuppose the independence of the predictors and time-varying characteristics of customer behavior. These premises are hardly feasible in contemporary online shopping conditions where purchase behavior varies with frequency highs and lows, use of omnichannels, and timely decision-making [1]. By contrast, AI-based strategies exploit the interaction of features and nonlinear dependencies by learning in an iterative manner, which allows them to make more accurate lifetime value and churn forecasts. Results as they are currently exemplified support a set of prior results [5], who say that behavioral and transactional heterogeneity is the way to go to increase predictive value and marketing ROI in CLV models.

One of the contributions of this research is proving that model interpretability can be achieved using SHAP (Shapley Additive Explanations) analysis. Whereas most AI models are black boxes, SHAP can be interpreted freely and makes it possible to assign contributions to the results of each predictor [15]. Marketing practitioners must have this interpretability, which enables them to convert the insights of prediction into actionable strategies. An example is that SHAP analysis provided primarily purchase frequency, recency, and average order value as the key determinants of CLV- information that can directly be applied to customer segmentation, loyalty program design, and personal retention campaigns [22].

These findings have various implications as far as their management is concerned. First, AI-based CLV systems allow marketers to focus on high-value customers to retain them and distribute acquisition budgets more efficiently. Decision-making based on such segmentation contributes to the optimization of the resources and improved marketing return on investment (ROI) [20]. Second, by combining churn modeling and prediction, it is possible to proactively engage at-risk customers before defection, as eventual profitability drives relationship-based marketing approaches [25]. Lastly, gradient boosting and neural models are scalable; businesses can use predictive analytics on huge amounts of customer data to process in real-time, a feature that is invaluable in a competitive e-commerce and subscription market [13].

Although AI-based CLV models have benefits, they have a few limitations. Possible model bias, overfitting, and issues with data privacy are the most conspicuous ones. The phenomenon of overfitting can arise when models acquire noise rather than actual patterns of behavior, particularly in small and imbalanced samples [23]. To counter this, cross-validation, regularization, and hyperparameter tuning techniques were used. Additionally, prediction can be biased by the underrepresentation of particular groups of people and behaviors, causing the underrepresentation of specific groups and thereby affecting the fairness of marketing behavior [11]. Data security and ethical consideration within the framework of such laws as the General Data Protection Regulation (GDPR) are also instrumental when using AI in processing sensitive customer data.

This study is part of more general trends in AI-based business analytics in which predictive models are currently being applied to finance, healthcare, and logistics to optimize lifetime outcomes [24]. The role played by AI as a strategic facilitator of decision intelligence in any industry is only enhanced by the interpretability and predictive power exhibited here.

PRACTICAL IMPLICATIONS AND RECOMMENDATIONS

The use of AI-guided CLV insights is helping marketing teams to be data-focused towards customer relationship management. Predictive modeling can help marketers direct the allocation of promotional budgets within high-value segments to allocate the retention budget to the most promising clients in terms of their lifetime profitability [19]. This accuracy lowers the cost of wastage in the acquisition cost and also improves marketing return on investment (ROI) [20].

By incorporating AI-driven analytics into Customer Relationship Management (CRM) and marketing automation systems, organizations will be able to dynamically tailor their campaigns by matching communications and offers with estimated customer behavior [26]. Salesforce and HubSpot platforms are becoming more controlled by machine learning modules that capture churn warning signals and dynamic segmentation [27].

In addition, predictive CLV modeling can aid in designing loyalty programs at the tiers of loyalty, advanced retention, and customized pricing (to enhance customer interaction) [25]. Display of transparency in models should be used to enhance business results so that performances are ethical and compliant with data, using interpretable AI components like SHAP [28]. In general, AI-based CLV models offer practical intelligence to use in the allocation of resources strategically, retention, and growth of customer value sustainably in the competitive market environment

CONCLUSION

This report has proven that AI-based CLV and churn prediction models are far more accurate, flexible, and explainable than the more traditional approaches that rely on statistics. After using ensemble learning tools like XGBoost and combining them with the SHAP-based explanations, the organizational members can not only predict the value of customers with the required level of accuracy but also identify the main drivers of behavior affecting the retention.

The results confirm that predictive analytics can shift the marketing strategy to proactive rather than reactive and make the marketing strategy engaging the target audience personally while allocating the budget and boosting ROI. The application of AI to the CRM and marketing automation solution offers a scalable route for operationalizing insights on a real-time basis to bridge the data science and managerial decision-making procedure.

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